

### TECHNOLOGY OF METALLIC MATERIALS 2025-206 -ASSIGNMENT N.3

Closed book, closed notes, no help from other students.  
If no attempt, search through the notes or the Mangonon book  
All justifications should use materials engineering principles.

For Questions 1- 4, match ALL correct answers a)-n) with the statements (answers can be used more than once and multiple answers are possible).

a) austenite      b) bainite      m) martensite      p) pearlite      n) none of these

- p,b 1. Possible microstructure in a 1080 steel at room temperature  
b,m 2. Microstructures which are possible following the rapid cooling (quenching) of pearlite  
m 3. The most brittle microstructure  
a 4. Single phase with Carbon in solid solution with Fe

For Questions 5-7 match ALL correct answers a) – n) with the statements. Recall that in both Quench and Temper of steel and the Age (Precipitation) Hardening of Aluminum there is a Heat, Quench, and Reheat process.

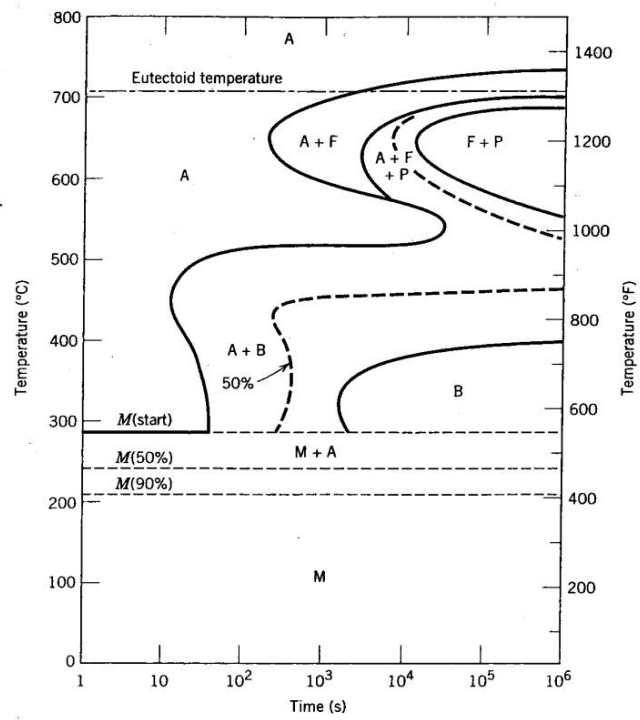
a) age (precipitation) hardening      q) quench and temper      n) neither

- q 5. Martensite forms during quenching  
a 6. Cold rolling or other plastic deformation is sometimes done between the quench and reheat steps.  
q 7. A phase change occurs upon quenching  
n 8. A second phase precipitates during the reheat process.  
a 9. Maximum strength peaks sometime after start of final reheating

10. For precipitation hardening (or age hardening) select the truest statement

- a) The formation of the  $\theta''$  phase, a precipitate coherent (lined-up with) with the Al matrix, means the alloy is over-aged.  
☒ b) The coherent precipitate, ( $\theta''$  phase) is finely dispersed and gives the best dislocation locking capability.  
c) The welding of precipitation hardened Al components should be delayed until after the ageing process is complete.  
d) The  $\theta$  phase precipitates, which form after the  $\theta''$  phase, are not coherent with the Al matrix and give the best dislocation locking capability  
e) The aging step requires heating to above the eutectic temperature.

**FIGURE 10.15**  
Isothermal transformation diagram for an alloy steel (type 4340): A, austenite; B, bainite; P, pearlite; M, martensite; F, proeutectoid ferrite. (Adapted from H. Boyer, Editor, *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, American Society for Metals, 1977, p. 181.)



11. Use the Isothermal Transformation Diagram for a 4340 hypoeutectoid steel to answer the following questions. To describe a microstructure, give the microconstituent or microconstituents present, and the relative amounts of each. For full credit, describe any pearlite that you find as "fine" or "coarse." Assume that we begin with A, austenite.

- a) Describe the microstructure which results from a rapid (taking less than 5 sec) quench to 100 C.

The result will be 100% martensite since the path went straight down, missing pearlite and bainite regions

- b) Describe the microstructure which results from a rapid quench to 300 C, holding for  $10^4$  seconds and then quenching rapidly to 100 C.

Bainite (almost all austenite transforms into it)

- c) Describe the microstructure which would result from a rapid quench to 400 C, holding for 1000 seconds, and then quenching rapidly to 100 C.

Around 50/50 between bainite and austenite

12. If a 100% Bainitic microstructure were desired in the 4340 steel described in the above TTT diagram, a reasonable procedure would be to

- a) Heat the steel to 300C from room temperature, start your timing and hold it there for  $10^4$  seconds, then quench back to room temperature.
- b) Heat the steel to 300C from room temperature, start your timing and hold it there for  $10^6$  seconds, then quench back to room temperature.
- c) Heat the steel to 800C from room temperature, quench to 450 C, hold for  $10^6$  seconds and then quench back to room temperature.
- d) Heat the steel to 800 C from room temperature, quench to 300 C and hold it there for about 50 seconds, then quench back to room temperature.
- ☒ e) Heat the steel to 800 C from room temperature, quench to 300 C and hold it there for about  $10^4$  seconds, then quench back to room temperature.

13. Assume that the Bainite was reheated to 600 C and left for  $10^4$  seconds. Then a quench to 100 C is performed. The microstructure which results is

- a) Fine Pearlite
- ☒ b) Proeutectoid Ferrite and Fine Pearlite
- c) Proeutectoid Ferrite, Fine Pearlite, and Martensite
- d) Martensite
- e) Bainite

14. A 2.5 inch diameter bar made of 8640 steel is quenched in mildly agitated water. Cooling rates as a function of position in the bar are shown in Figure A. Use the hardenability curves in Figure B to predict the hardness, HRC, at
- (a) the surface of the bar 55
- (b) the center of the bar 35

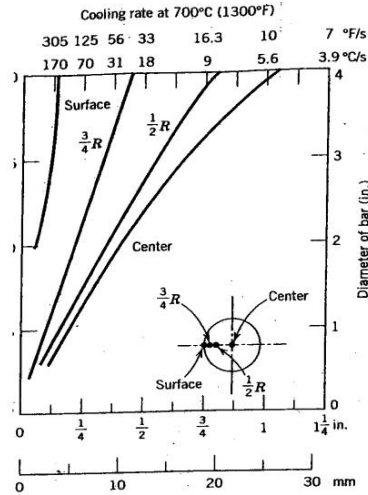


Figure A.

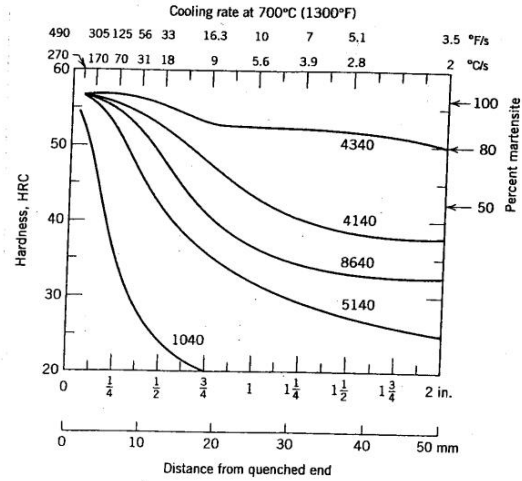
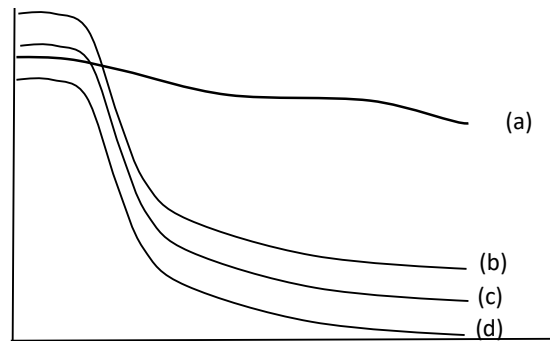


Figure B.

For the hardenability curves shown, the four alloys present are 1020, 1040, 1060 and 4130.

Match the best answer a) –d) with the statement below

- d? 15. 1020 steel
- a 16. Highest hardenability



### 17. Define Hardenability

The ability of a steel or iron alloy to harden when quenched.